

LOG ANALYSIS WITH WAZUH and SURICATA

EDR Fundamentals with WAZUH and SURICATA.

Hands-On EDR with Wazuh and Suricata: 5 days of log analysis From Setup to Threat Detection.

Tools : UBUNTU, KALI, HYDRA, WAZUH, SURICATA, Nmap, Emerging Threat Rules, CVE Databases [NVD & CANONICAL SECURITY TRACKER].

Scenarios Investigated:

Day 26: Set up Wazuh server and agent to establish endpoint monitoring.

Day 27: Detected unauthorized file changes using Wazuh File Integrity Monitoring.

Day 28: Simulated SSH brute-force with Hydra and confirmed Wazuh alerts.

Day 29: Integrated Suricata with Wazuh and caught an Nmap port scan.

Day 30: Used Wazuh Vulnerability Detector to identify system CVEs.

Day 26: Wazuh Setup

Objective

Deploy a working Wazuh Server and onboard agents to establish centralized log collection and monitoring.

Tasks Performed:

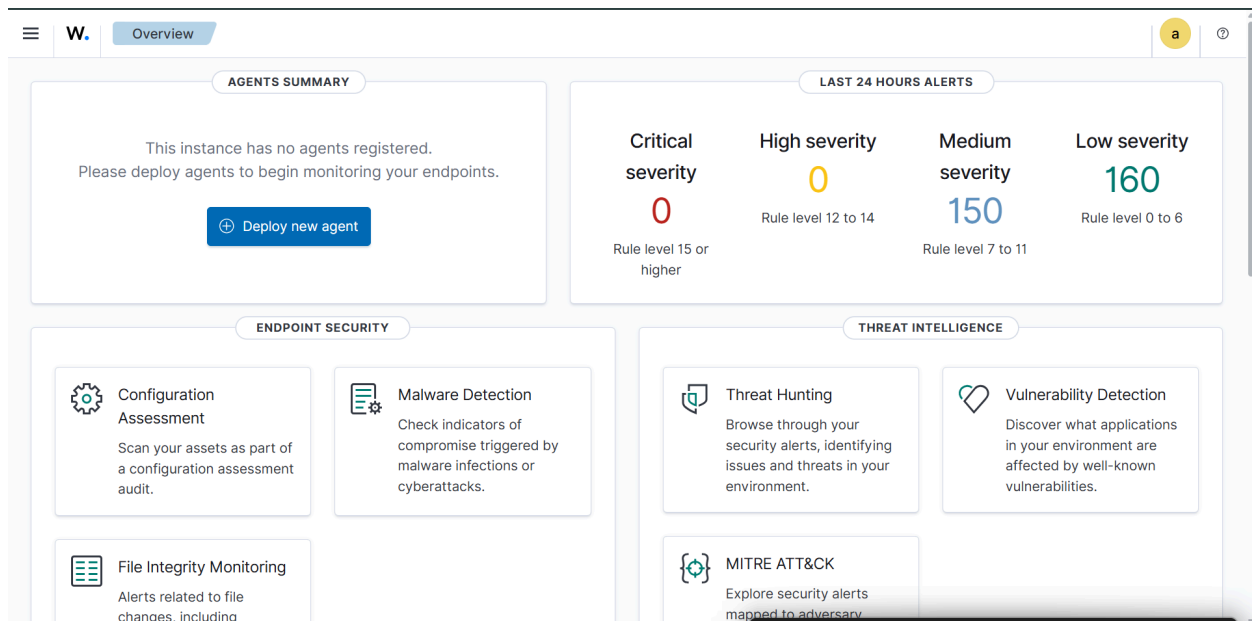
Installed the Wazuh server using the quick-start installation script.

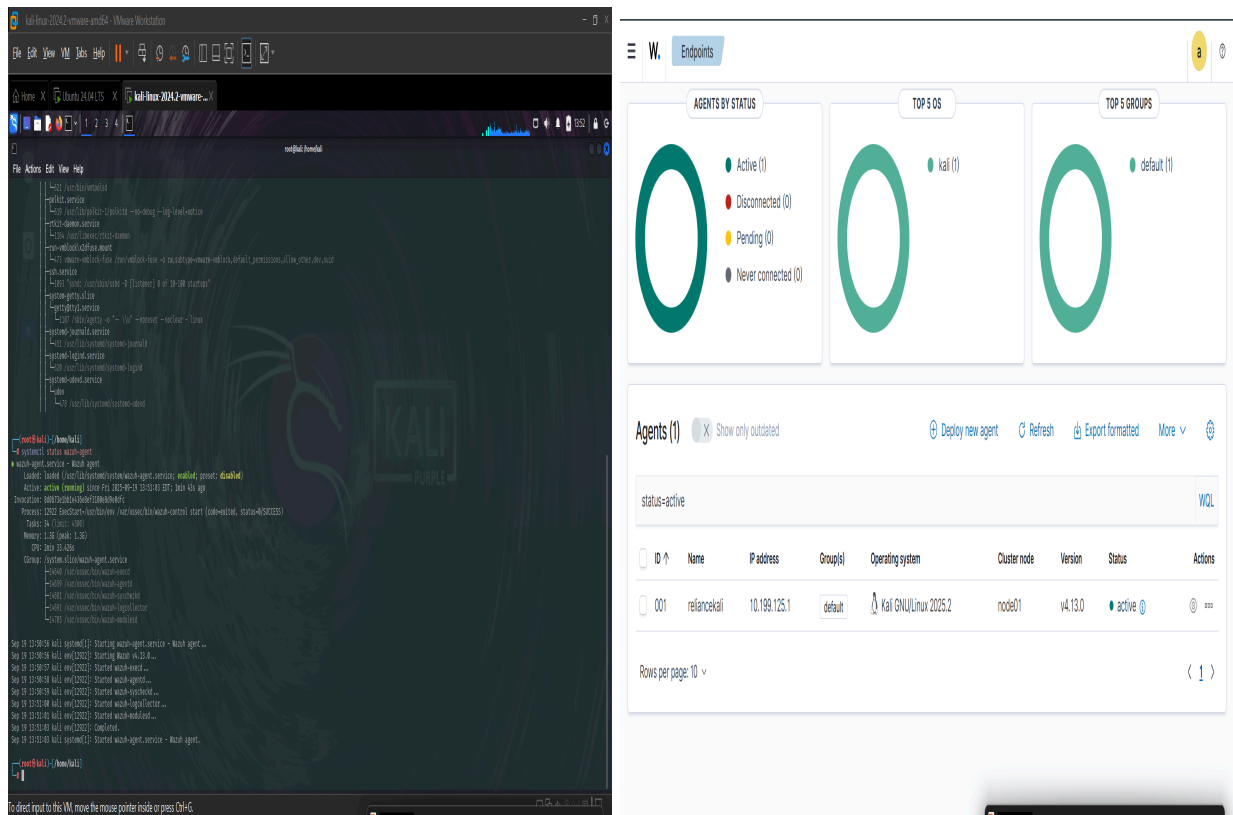
Accessed the Wazuh Dashboard through the web interface and logged in with admin credentials.

Installed the Wazuh Agent on a separate Ubuntu endpoint.

Configured the agent to point to the Wazuh server's IP and enabled the service.

Verified agent status on the dashboard, confirming successful server-agent communication.





Gained practical experience in setting up the Wazuh environment, confirming how agents feed system events into the manager, which forms the foundation for advanced threat detection and endpoint visibility.

Day 27: File Integrity Monitoring (FIM)

Objective:

Configure and test File Integrity Monitoring (FIM) to detect unauthorized file access or modifications.

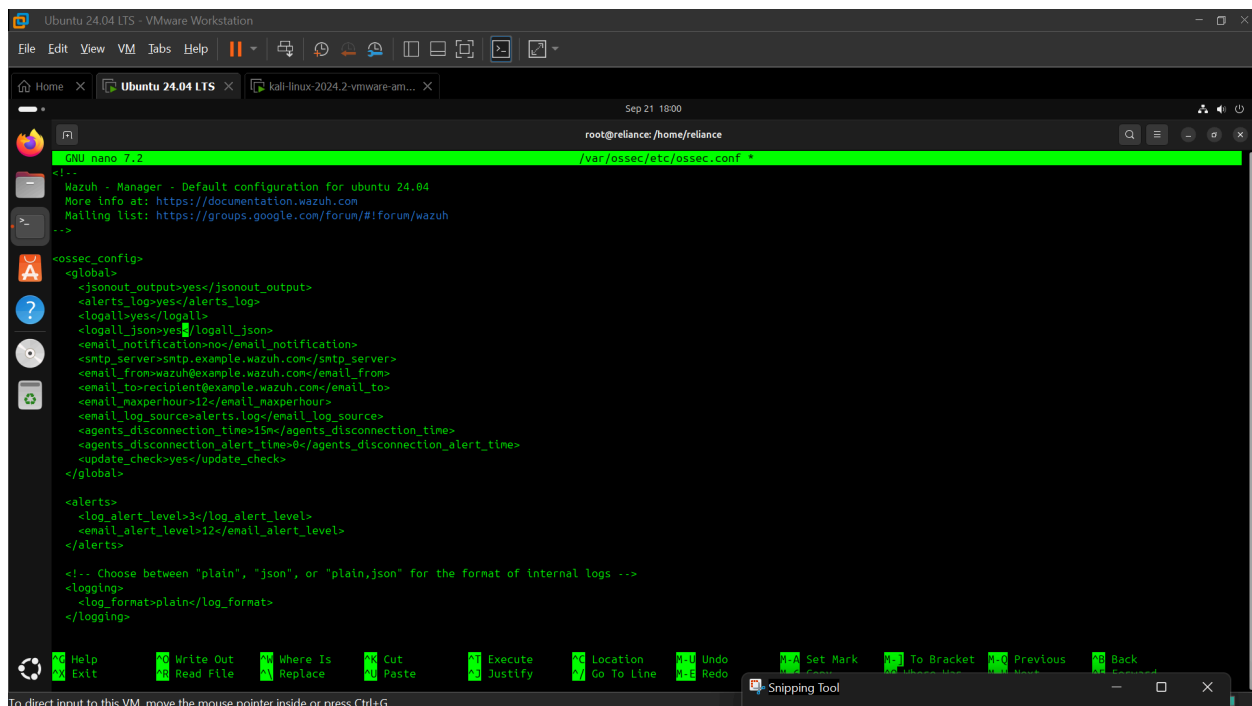
Tasks Performed:

Configured the Wazuh Manager to monitor C:\Users\Public\Documents on a Windows agent.

Restarted the Wazuh Manager to apply the new FIM configuration.

Simulated malicious behavior by creating, modifying, and deleting a file named important.txt.

Used the Wazuh Dashboard to track and confirm alerts generated for each file action (creation, modification, deletion).



```
Ubuntu 24.04 LTS - VMware Workstation
File Edit View VM Tabs Help
Home x Ubuntu 24.04 LTS x kali-linux-2024.2-vmware-am... x
Sep 21 18:00
root@reliance: /home/reliance
GNU nano 7.2 /var/ossec/etc/ossec.conf
<!--
Wazuh - Manager - Default configuration for ubuntu 24.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>yes</logall>
    <logall_json>yes</logall_json>
    <email_notification>no</email_notification>
    <smtp_server>smtp.example.wazuh.com</smtp_server>
    <email_from>wazuh@example.wazuh.com</email_from>
    <email_to>recipient@example.wazuh.com</email_to>
    <email_maxperhour>12</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>15m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
    <update_check>yes</update_check>
  </global>

  <alerts>
    <log_alert_level>3</log_alert_level>
    <email_alert_level>12</email_alert_level>
  </alerts>

  <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
  <logging>
    <log_format>plain</log_format>
  </logging>

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark To Bracket Previous Back
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

File Integrity M...

reliancekali

/boot/System.map-6.6.15-amd64

/boot/config-6.12.33+kali-amd64

/boot/config-6.6.15-amd64

/boot/grub/fonts/unicode.pf2

/boot/grub/grub.cfg

/boot/grub/grubenv

/boot/grub/i386-pc/915resolution.mod

/boot/grub/i386-pc/acpi.mod

/boot/grub/i386-pc/adler32.mod

/boot/grub/i386-pc/affs.mod

/boot/grub/i386-pc/afs.mod

/boot/grub/i386-pc/afsplitter.mod

/boot/grub/i386-pc/ahci.mod

Rows per page: 15

/boot/grub/i386-pc/afsplitter.mod

Details

Last analysis

Sep 19, 2025 @ 18:51:33.000

Last modified

Jul 10, 2025 @ 11:56:52.000

User

root

User ID

0

Group

root

Group ID

0

Size

1.47 KB

Inode

1704518

MD5

0c0bf8eb8d13b09c2fd6d966e8008ecf

SHA1

f5a38c60cd9a1bf153c44c23de180f8c9f1078

SHA256

70a47237eb6c4d2eda4e9eb107f6b05604c100fda126392446415720a821a74e

Permissions

rw-r--r--

Recent events

Search

DQL

Last 24

kali-linux-2024.2-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Home

Ubuntu 24.04 LTS

kali-linux-2024.2-vmware-...

1 2 3 4

13:13

kali@kali -

var/ossec/etc/ossec.conf

```

<!-- Database synchronization settings -->
<synchronization>
  <max_eps>10</max_eps>
</synchronization>
</wodle>

<!-- File Integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,<!-- Directories to ignore -->
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/passwd</ignore>
  <ignore>/etc/mail/statistics</ignore>
  <ignore>/etc/random-seed</ignore>
  <ignore>/etc/random-seed</ignore>
  <ignore>/etc/adjtime</ignore>
  <ignore>/etc/httpd/logs</ignore>
  <ignore>/etc/utmp</ignore>
  <ignore>/etc/wtmp</ignore>
  <ignore>/etc/cups/certs</ignore>
  <ignore>/etc/udev/rules</ignore>

```

Help Exit

Write Out Read File

Where Is Replace Paste

Execute Justify

Location Go To Line

Undo Redo

Set Mark Copy

To Bracket Where Was

Previous Next

Back Forward

Prev Word Next Word

Home End

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Key Takeaways:

Learned how FIM provides immediate visibility into sensitive file activity, a critical security control for detecting insider threats, tampering, or malware behavior.

Day 28: SSH Brute Force Detection

Objective:

Simulate and detect brute-force attempts against SSH services using Wazuh rules and logs.

Tasks Performed:

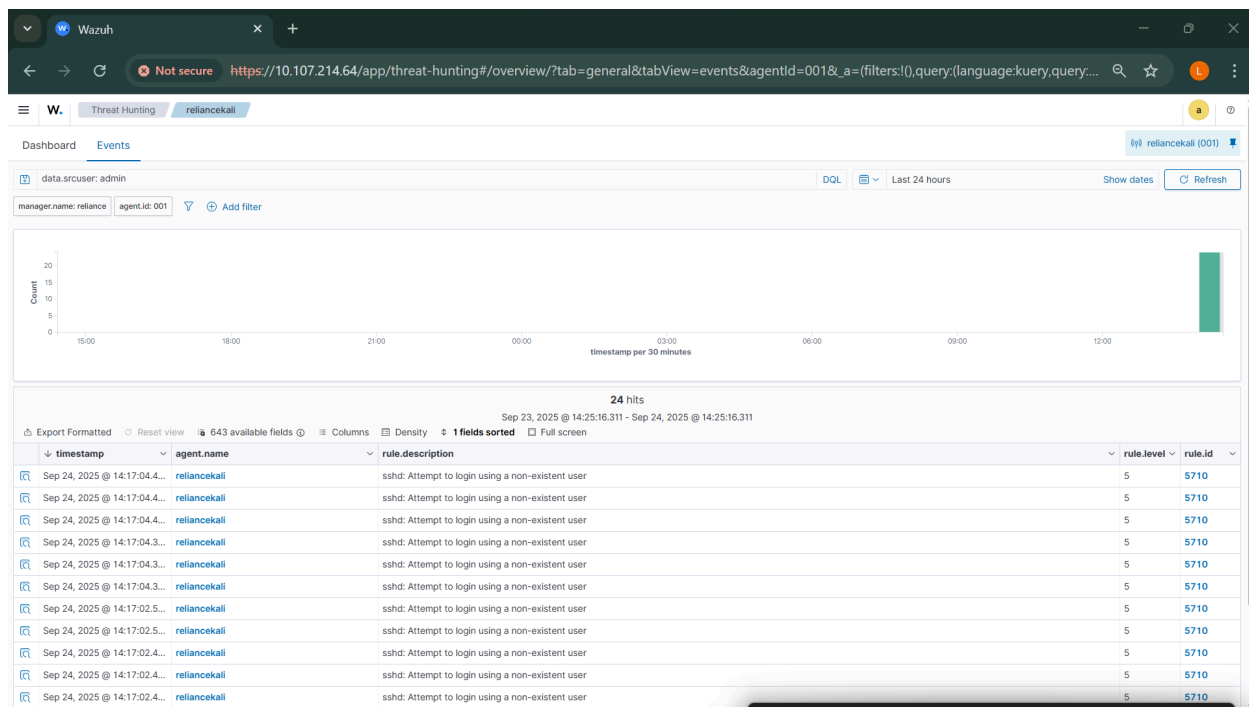
Installed and configured Hydra on Kali Linux as the attacking machine.

Created a simple password list for brute-force attempts.

Ran Hydra with SSH brute-force parameters against the Ubuntu server.

Monitored /var/log/auth.log to verify repeated failed login attempts.

Queried Wazuh logs (alerts.json) and confirmed alerts under rule IDs 5551 and 5712 indicating brute-force activity.



Wazuh Threat Hunting

Dashboard Events

Wazuh Alerts - admin - agentId: 001 & a=(filters:[];query:(language:kuery.query...))

Document Details

Table JSON

```
{
  "index": "wazuh-alerts-4.x-2025.09.24",
  "agent_id": "001",
  "agent_ip": "10.107.214.1",
  "agent_name": "reliansekali",
  "data_srcip": "10.107.214.1",
  "data_username": "admin",
  "decoder_name": "sshd",
  "decoder_parent": "sshd",
  "full_log": "Sep 24 13:17:02 kali sshd-session[5489]: Failed password for invalid user admin from 10.107.214.1 port 48916 sshd",
  "id": "175871824.37907",
  "input_type": "log",
  "location": "journal2",
  "manager_name": "reliansec",
  "predecoder_hostname": "kali",
  "predecoder_program_name": "sshd-session",
  "predecoder_timestamp": "Sep 24 13:17:02",
  "rule_description": "sshd: Attempt to login using a non-existent user",
  "rule_firewalltime": "9",
  "rule_gdb": "IX.35.7.4, IX.32.3",
  "rule_gid": "7.1",
  "rule_groups": "syslog, sshd, authentication, failed, invalid_login",
  "rule_hipaa": "164.312.0",
  "rule_id": "5719",
  "rule_level": "5",
  "rule-met1": "false",
  "rule-met13": "false",
  "rule-actre-id": "7110-001 71021-004",
  "rule-actre-tactic": "Credential Access, Lateral Movement",
  "rule-actre-technique": "Password Guessing, SSH",
  "rule-nist_800_33": "Au.14, Ac.7, Au.6",
  "rule-nist_800_33": "10.2.4, 10.2.5, 10.6.1",
  "rule-see": "CDE.1, CDE.8, OCT.2, OCT.3",
  "timestamp": "Sep 24, 2025 @ 14:17:04.486"
}
```

kali-linux-2024.2-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Home x Ubuntu 24.04 LTS x kali-linux-2024.2-vmware-... x

root@kali:/home/kali

```
File Actions Edit View Help

[kali@kali]~$ sudo su
[sudo] password for kali:
[root@kali]~/home/kali# hydra -l admin -P password.txt 10.107.214.1 ssh
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-24 09:16:58
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 11 tasks per 1 server, overall 11 tasks, 11 login tries (1:1/p:11), ~1 try per task
[DATA] attacking ssh://10.107.214.1:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-24 09:17:02

[root@kali]~/home/kali#
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Key Takeaways:

Understood how Wazuh leverages system logs and correlation rules to detect repeated failed logins, providing early warning signs of brute-force attacks that could escalate to unauthorized access if unchecked.

Day 29: Suricata + Wazuh Integration

Objective:

Enhance detection of suspicious network traffic by integrating Suricata IDS with Wazuh SIEM.

Tasks Performed:

Installed Suricata on the Ubuntu agent and downloaded Emerging Threats rules.

Configured Suricata with a custom rules directory and set HOME_NET to the host IP.

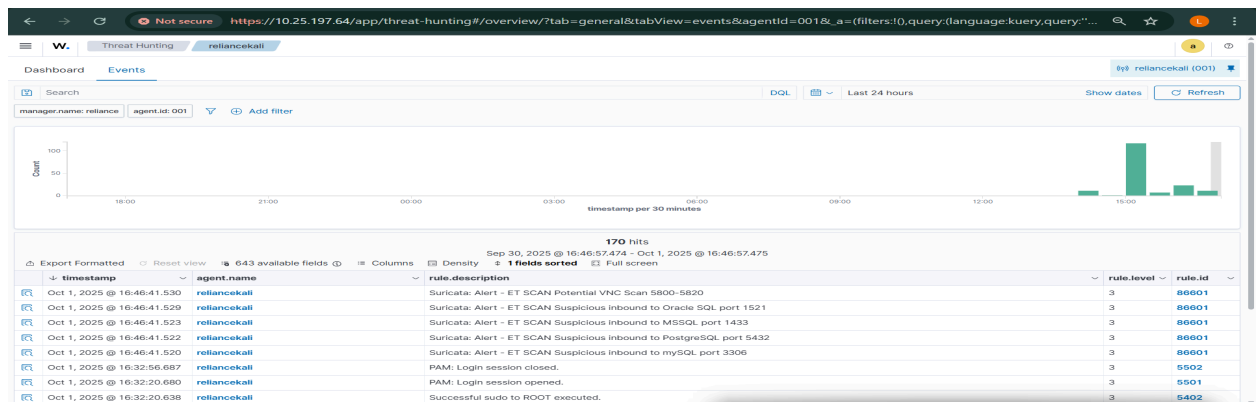
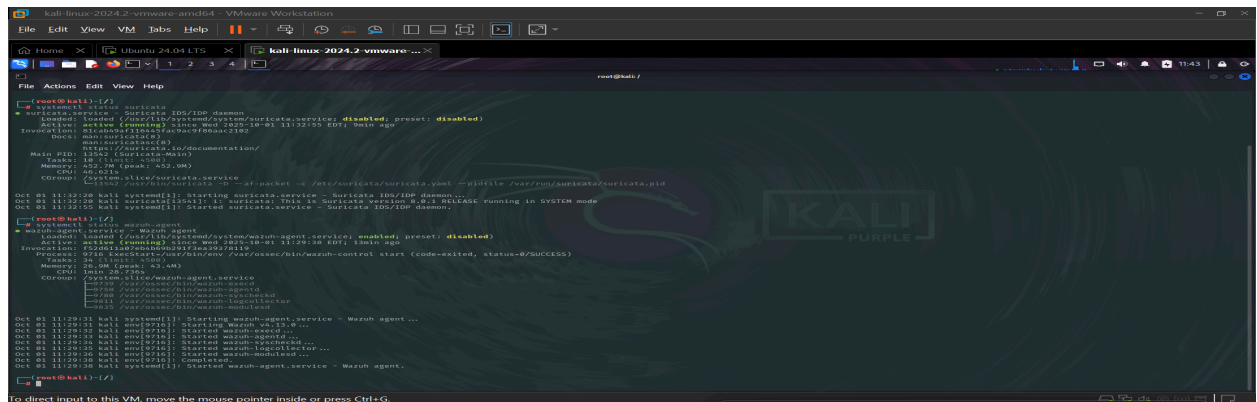
Enabled JSON logging (eve.json) for Suricata alerts.

Updated Wazuh Agent config to ingest Suricata's eve.json logs.

Launched a simulated port scan from Kali using Nmap (nmap -sS -T4).

Confirmed Suricata flagged the scan as ET SCAN Nmap Synchronous Scan and saw the same alert normalized in the Wazuh Dashboard.

The image shows a Kali Linux virtual machine environment. The top of the window displays the title bar 'kali-linux-2024.2-vmware-amd64 - VMware Workstation'. Below the title bar is a menu bar with 'File', 'Edit', 'View', 'VM', 'Tabs', and 'Help'. The main window area shows a terminal window with the prompt 'kali@kali: /etc/suricata/rules'. The terminal content lists various rule files in the directory, including 'rules/emerging-info.rules', 'rules/emerging-j3a.rules', 'rules/emerging-malware.rules', 'rules/emerging-mass.rules', 'rules/emerging-mobile_malware.rules', 'rules/emerging-metasploit.rules', 'rules/emerging-p2p.rules', 'rules/emerging-phishing.rules', 'rules/emerging-policy.rules', 'rules/emerging-pop3.rules', 'rules/emerging-rpc.rules', 'rules/emerging-rpc.rules', 'rules/emerging-scan.rules', 'rules/emerging-scan.rules', 'rules/emerging-shellcode.rules', 'rules/emerging-smtp.rules', 'rules/emerging-smtp.rules', 'rules/emerging-sql.rules', 'rules/emerging-telnet.rules', 'rules/emerging-tftp.rules', 'rules/emerging-user_agents.rules', 'rules/emerging-voip.rules', 'rules/emerging-web_client.rules', 'rules/emerging-web_server.rules', 'rules/emerging-web_specific_apps.rules', 'rules/emerging-worm.rules', 'rules/gpl-2.0.txt', 'rules/isp-mgmp.rules', 'rules/suricata-3.5-enhanced-open.txt', 'rules/threatview_CS_C2.rules', 'rules/tor.rules', 'rules/tor.rules'. The terminal also shows the command 'cd /etc/suricata/rules' and a list of rule files in a grid format. The background of the terminal window features the Kali Linux logo and the text 'KALI PURPLE'.

A screenshot of the Wazuh Threat Hunting interface showing document details. The top bar shows 'Threat Hunting' and 'reliancekall'. Below it, there's a 'Dashboard' and 'Events' tab. A search bar contains 'manager.name:reliancekall agent.id:001'. A bar chart shows 'hits' over time, with a peak around 15:00. Below the chart, a table lists 170 hits. The table has columns for timestamp, agent.name, rule.description, rule.level, and rule.id. The rules are related to Suricata alerts and PAM login sessions. The 'Document Details' panel on the right shows fields like data.flow.src_ip, data.flow.src_port, data.flow.start, data.flow.id, data.in_iface, data.ip_v, data.pkt_src, data.proto, data.src_ip, data.src_port, data.timestamp, r.decoder.name, r.id, r.input.type, r.location, r.manager.name, r.rule.description, r.rule.firedtimes, r.rule.groups, r.rule.id, r.rule.level, r.rule.mail, and timestamp.

Key Takeaways:

Demonstrated how IDS (Suricata) and SIEM (Wazuh) integration provides layered defense: Suricata detects network intrusions while Wazuh centralizes, normalizes, and correlates alerts for SOC analysts to act upon.

Day 30: Vulnerability Detection

Objective:

Identify endpoint software vulnerabilities using Wazuh's built-in Vulnerability Detector module.

Tasks Performed:

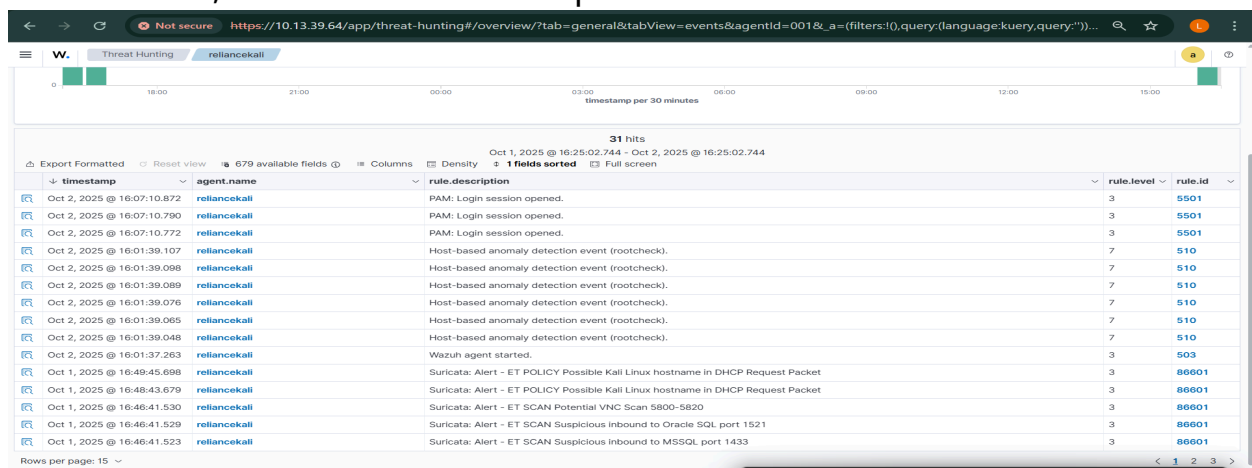
Enabled and configured the vulnerability detector in Wazuh's ossec.conf.

Restarted the Wazuh Manager and confirmed a clean startup.

Allowed the module to scan installed software packages and cross-reference them with the NVD and Canonical security tracker.

Reviewed alerts in the Wazuh Dashboard, filtering by severity and rule group.

Examined vulnerability alerts showing CVE IDs, affected package names, CVSS scores, and recommended patches.



timestamp	agent.name	rule.description	rule.level	rule.id
Oct 2, 2025 @ 16:07:10.872	reliancekali	PAM: Login session opened.	3	5501
Oct 2, 2025 @ 16:07:10.790	reliancekali	PAM: Login session opened.	3	5501
Oct 2, 2025 @ 16:07:10.772	reliancekali	PAM: Login session opened.	3	5501
Oct 2, 2025 @ 16:01:39.107	reliancekali	Host-based anomaly detection event (rootcheck).	7	510
Oct 2, 2025 @ 16:01:39.098	reliancekali	Host-based anomaly detection event (rootcheck).	7	510
Oct 2, 2025 @ 16:01:39.089	reliancekali	Host-based anomaly detection event (rootcheck).	7	510
Oct 2, 2025 @ 16:01:39.076	reliancekali	Host-based anomaly detection event (rootcheck).	7	510
Oct 2, 2025 @ 16:01:39.065	reliancekali	Host-based anomaly detection event (rootcheck).	7	510
Oct 2, 2025 @ 16:01:39.048	reliancekali	Host-based anomaly detection event (rootcheck).	7	510
Oct 2, 2025 @ 16:01:37.263	reliancekali	Wazuh agent started.	3	503
Oct 1, 2025 @ 16:49:45.698	reliancekali	Suricata: Alert - ET POLICY Possible Kali Linux hostname in DHCP Request Packet	3	86601
Oct 1, 2025 @ 16:48:43.679	reliancekali	Suricata: Alert - ET POLICY Possible Kali Linux hostname in DHCP Request Packet	3	86601
Oct 1, 2025 @ 16:46:41.530	reliancekali	Suricata: Alert - ET SCAN Potential VNC Scan 5800-5820	3	86601
Oct 1, 2025 @ 16:46:41.529	reliancekali	Suricata: Alert - ET SCAN Suspicious Inbound to Oracle SQL port 1521	3	86601
Oct 1, 2025 @ 16:46:41.523	reliancekali	Suricata: Alert - ET SCAN Suspicious Inbound to MSSQL port 1433	3	86601

Key Takeaways:

Learned how Wazuh automates vulnerability detection and correlates security advisories with endpoint inventory, giving analysts actionable intelligence to prioritize patches and reduce attack surface.

SUMMARY

On Day 26, I deployed the Wazuh server and onboarded an Ubuntu agent, establishing the foundation for centralized endpoint visibility. This setup enabled log collection and monitoring, a critical first step in building an EDR workflow for SOC operations.

By Day 27, I configured File Integrity Monitoring (FIM) to detect unauthorized file changes on a Windows machine. Simulated file creation, modification, and deletion generated real-time alerts, showcasing how Wazuh helps detect tampering and insider threats.

On Day 28, I simulated an SSH brute-force attack using Hydra from a Kali machine. Wazuh correlated repeated failed logins into high-severity alerts, proving its ability to detect brute-force attempts and highlight credential-based threats.

Finally, on Day 29 and 30, I integrated Suricata with Wazuh to detect network scans and enabled Wazuh's Vulnerability Detector to flag outdated packages against CVE databases. These labs demonstrated how combining IDS, SIEM, and vulnerability management creates a layered defense strategy that strengthens endpoint security.